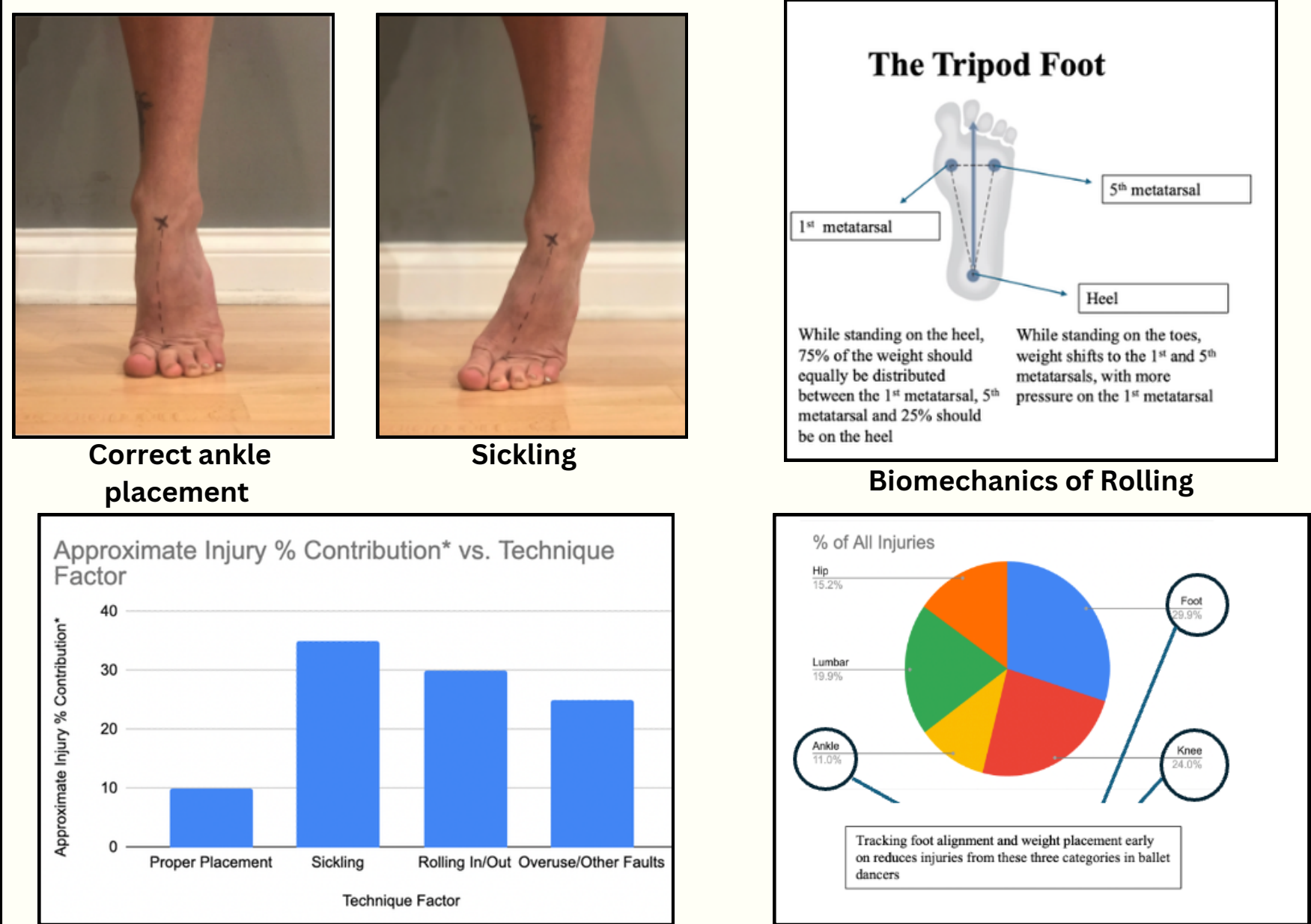


Problem Statement

During ballet pointe work, dancers must maintain foot and ankle alignments while supporting their body weight on a very small contact area. Small deviations from the optimal alignment like **sickling** (inversion and adduction of the rearfoot with lateral malleolus misalignment) or **rolling** (excessive load on the first or fifth metatarsal) alter how forces transmit through the foot-ankle complex, increasing the risk of *ankle sprains, tendonitis, stress fractures* and reduced postural stability.



Injury	Associated Ballet Biomechanics	Main Symptoms and Early Signs
Achilles Tendinitis	Poor alignment and excessive plantarflexion	Heel/ankle pain, stiffness and swelling
Ankle Sprains	Rolling and sickling	Sudden pain, swelling and reduced mobility
Stress Fractures	Poor Weight distribution	Persistent localized pain, swelling, tenderness
Cuboid Subluxation	Rolling Inwards and Outwards	Lateral foot pain, sense of instability

Limitations of Existing Methods

Diagnostic Solutions:

- Most identification of sickling and rolling relies heavily on teacher judgement and video recordings which are subjective
- Subtle/intermittent sickling and rolling often goes unnoticed especially during rapid movements and until injury develops

Current Sensor-Based Solutions:

- Clinical screenings are often conducted barefoot or in static positions whereas the issue occurs during dynamic movements
- Traditional biomechanics tools cannot be used inside pointe shoes and do not capture toe-box pressure asymmetry

Introduction

Engineering Problem: How can a lightweight, low power, in-shoe wearable system be engineered to accurately capture plantar pressure and inertial data in real time, enabling detection of misalignment and injury-related loading patters in dancers.

Research Question: To what extent can in shoe wearable systems detect unsafe weight distribution, alignment errors, and injury associated movement patterns in ballet dancers?

Finding a solution: EquiPointe is a non-invasive wearable embedded system that monitors plantar pressure and foot kinematics to detect unsafe loading and alignment patterns linked to injury risk in dancers.

- Research Objectives**
- Develop a low-profile, low-power in-shoe system integrating FSR and IMU sensors
 - Implement a real-time risk analysis algorithm for unsafe loading patterns
 - To extract and analyze foot kinematic parameters to quantify alignment, symmetry and rolling
 - Develop a system that is cost effective, easy to use and deployable at a larger scale

Potential User Risk	Possible Cause	Impact on user	Mitigation Strategy
Discomfort or prssure points	Sensors, wiring, or battery placement inside and outside the shoe	Foot pain, blisters or altered movement	Use ultra thin, flexible sensors; position components away from pressure zones; conduct short fit trials before use
Skin irritation	Prolonged contact with sensor materials or adhesives	Redness, irritation or an allergic reaction	Remove immediately if irritation occurs, exclusion of participants with pre-diagnosed skin conditions
Slipping or instability	Sensors shifting inside the shoe, unfamiliarity with the hardware	Loss of balance or falls	Secure sensors using a hot glue gun, taking appropriate measures to ensure they don't move; test stability during warm up movements

Methodology

Motion Tracking with MPU6050 & Force Detection with FSR402

3D Motion Measurement: The MPU6050 is a sensor that measures acceleration in three axes (A_x, A_y, A_z) and angular velocity in three axes (G_x, G_y, G_z) as the dancer moves.

Madgwick Filter: The filter minimizes the error between measured and expected gravitational field vectors using a gradient descent algorithm.

FSR402, Force Resistive Sensor: The Force Sensitive Resistor (FSR) measures the normal force F applied by the dancer's foot. Its resistance R decreases nonlinearly with increasing force, typically following $R \propto F^{-n}$.

Sensor Placement Strategy

Pointe-Shoe Sensor Placement

According to research by Manna et al and gait studies using foot mounted IMUs:

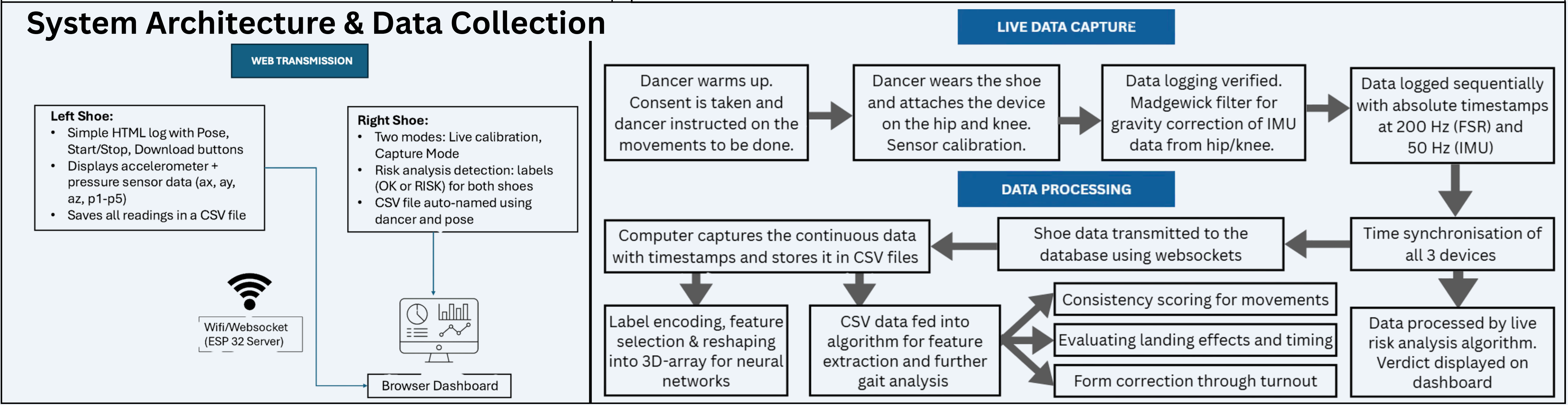
- FSR sensors are placed under the heel and below the 1st and 5th metatarsals.
- MPU6050 placed on the dorsum (top) of the foot

We perform **sensor-to-segment alignment** using gravity to define the vertical axis and functional knee motion to define the hinge axis, allowing accurate kinematics independent of sensor strapping.

Hip IMU sensor

Knee IMU sensor

Placement above the hip and knee allows estimation of joint angles



Data Analysis

Live Sickling Detection Methodology

Inputs:

IMU: Quaternion orientation $q = (q_0, q_1, q_2, q_3)$, angular velocity ω_x & vertical acceleration a_y .

FSRs: S_1 (toe-left) & S_2 (toe-right)

Roll angle γ (foot tilt) is obtained from the IMU quaternions:

$$\gamma = \tan^{-1} \left(\frac{2(q_0 q_1 + q_2 q_3)}{1 - (q_1^2 + q_2^2)} \right)$$

Toe pressure and imbalance are measured as:

$$P_{toe} = S_1 + S_2, \Delta_{toe} = \frac{S_1}{S_1 + S_2}$$

When large $|\gamma|$ (excessive inward/outward tilt), high $|\Delta_{toe}|$ (uneven toe loading) & high P_{toe} occur together, the system detects sickling.

Live Rolling Detection Methodology

Inputs:

Five FSR sensors: S_1, S_2, S_3, S_4, S_5

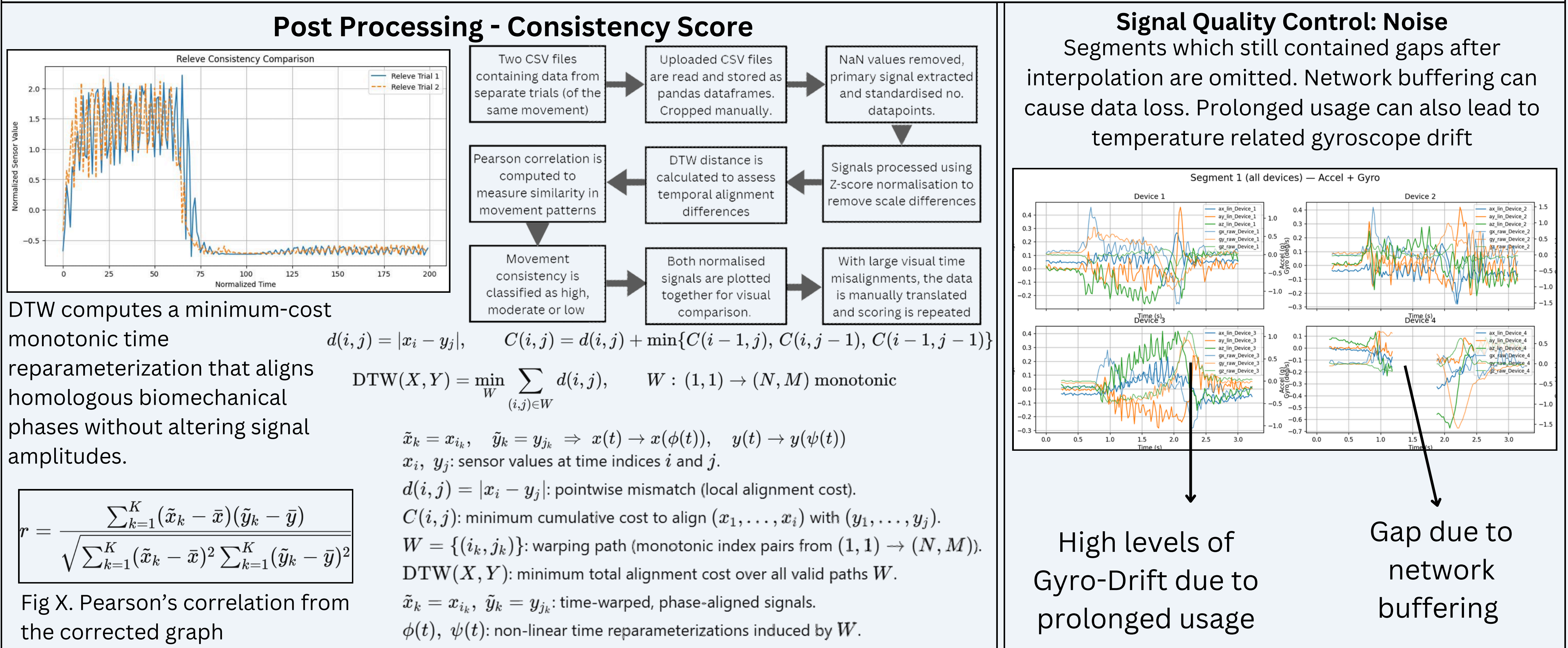
Medial and lateral pressures are defined as:

$$P_{medial} = S_1 + S_4, P_{lateral} = S_2 + S_5$$

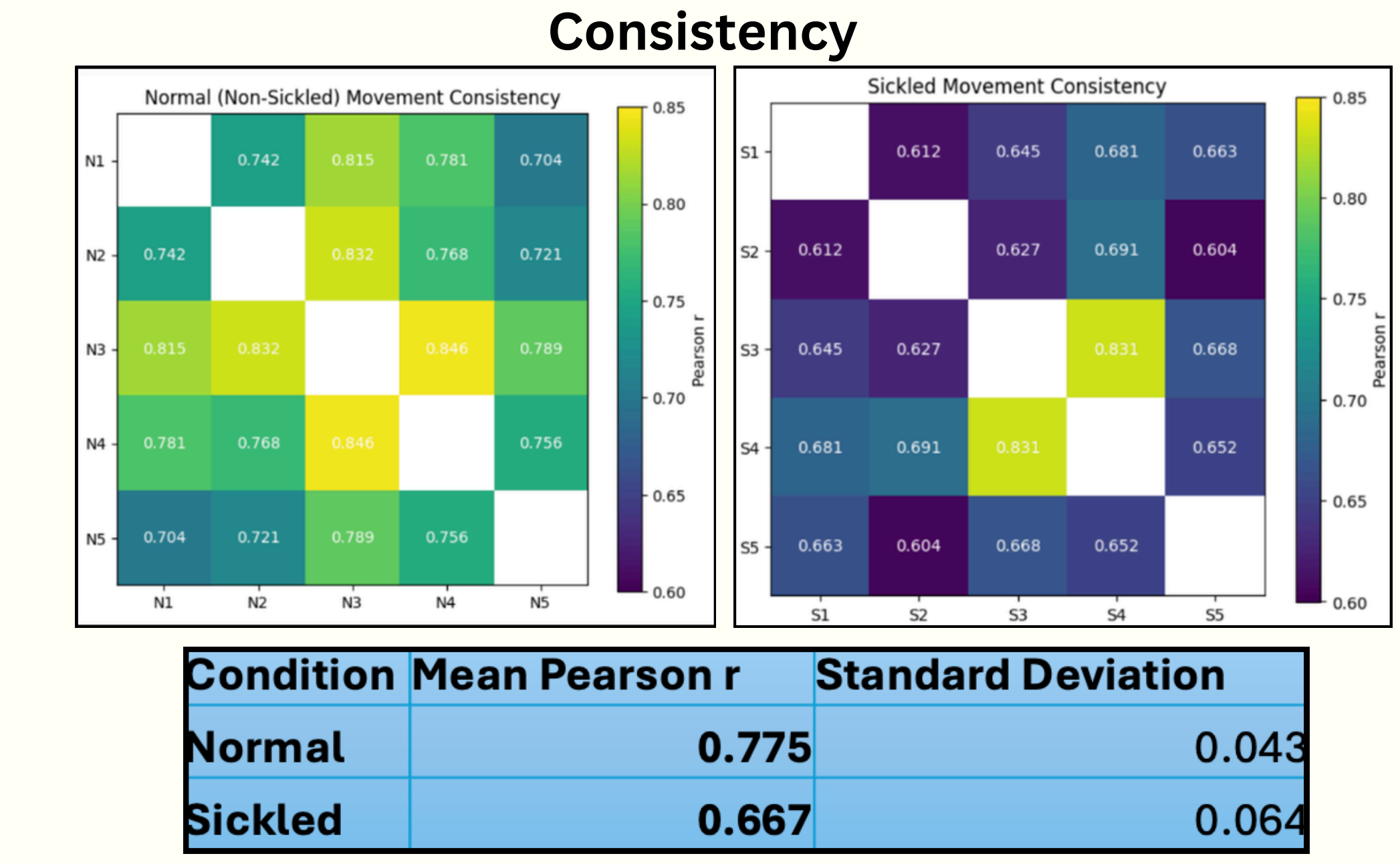
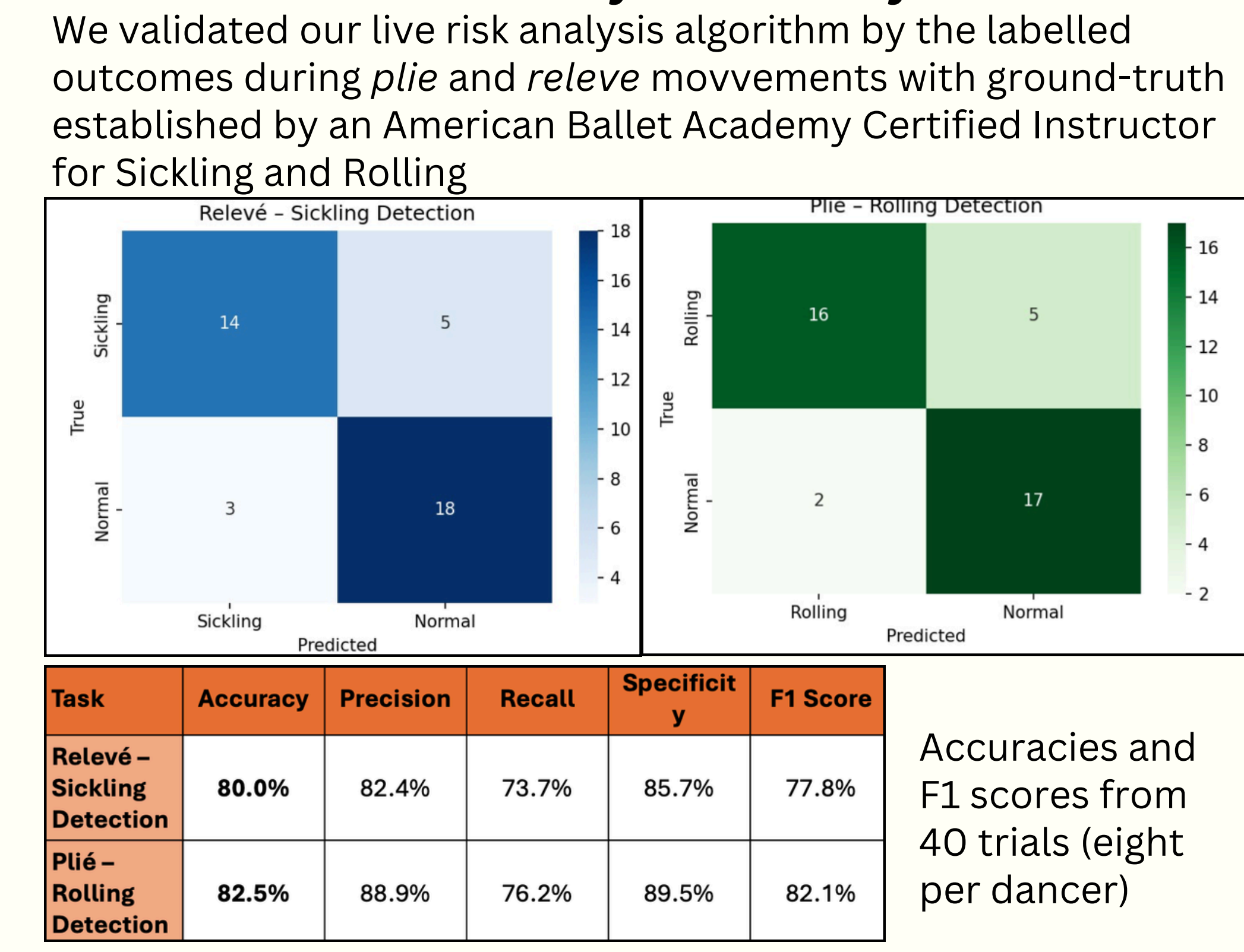
We extracted a feature Rolling Index (RI) which quantifies side imbalance using asymmetry in FSR measurements:

$$RI = \frac{P_{lateral} - P_{medial}}{P_{lateral} + P_{medial}}$$

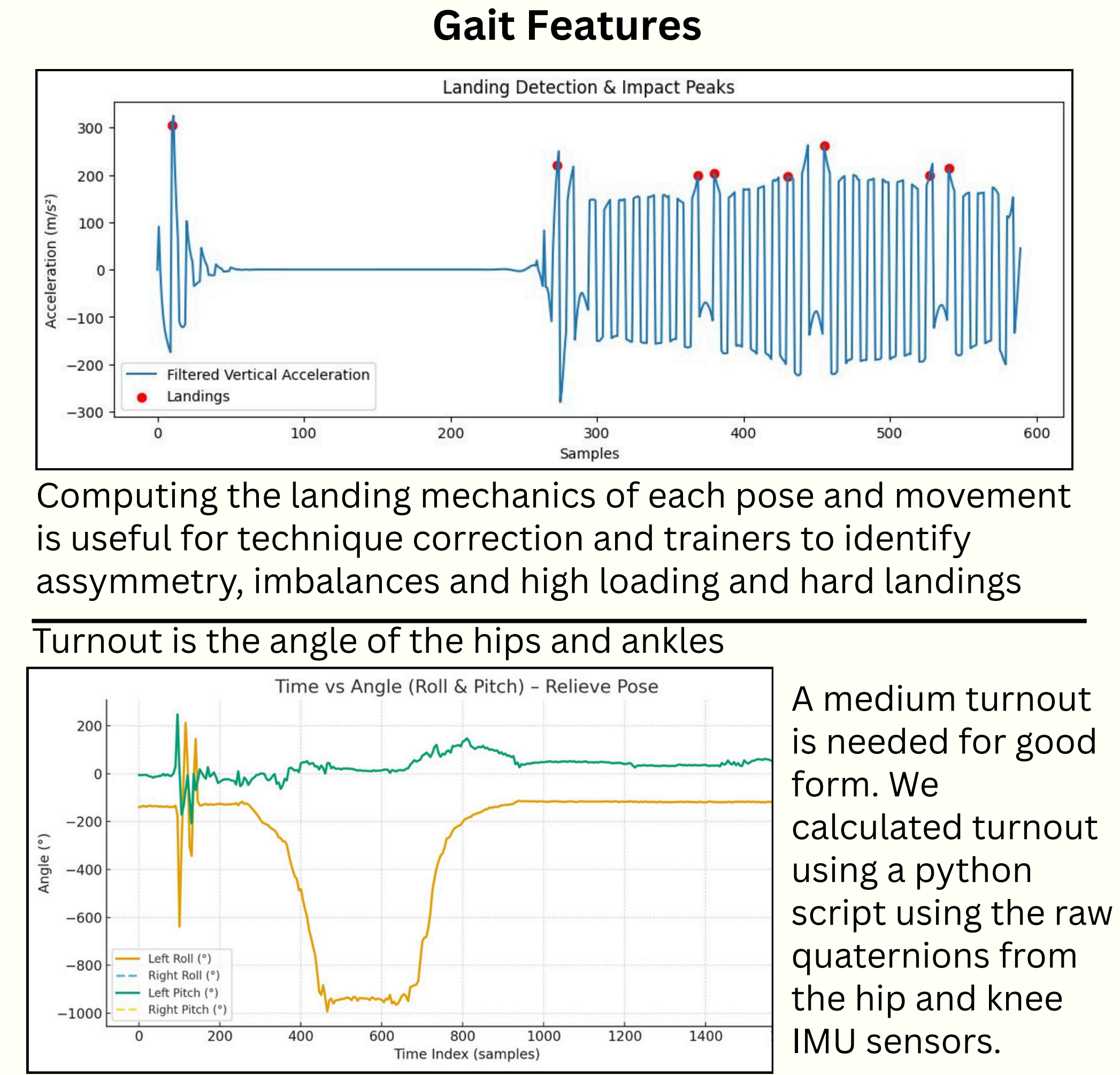
Emperically, we determined $-0.1 < RI < 0.1$ is balanced stance, $RI > 0.1$ is rightward roll & $RI < -0.1$ is leftward roll.



Results



From the mean pearson r, we conducted an independent samples t-test, obtaining $t \sim 4.15$ and $p \sim 0.002$ which indicates a statistically significant difference across groups. This shows incorrect technique directly leads to inconsistent movements.



Turnout is the angle of the hips and ankles

A medium turnout is needed for good form. We calculated turnout using a python script using the raw quaternions from the hip and knee IMU sensors.

- Future Scope**
- The same concept can be applied to all sports, specifically for running
 - Ground contact forces, loading symmetry, timing and movement cosistency are relevant across most sports
 - Sensor placement can be adapted for sports specific injuries
 - Consistency and deviation metrics can be applied to any movements, such as jumps, cuts and landings
 - Thresholds and risk models can be sport-specific, trained using data from healthy vs injured athletes in each sport

Key References

- Academy, U.S.S. (2019) Biomechanical Comparison of 'Dead' and 'New' Pointe Shoes in Female Professional Ballet Dancers, The Sport Journal. Available at: <https://thesportjournal.org/article/biomechanical-comparison-of-dead-and-new-pointe-shoes-in-female-professional-ballet-dancers/>.
- Engineering Meets Ballet: A Smart Solution for Safer Pointe Work - Johns Hopkins Whiting School of Engineering (2025) Johns Hopkins Whiting School of Engineering. Available at: <https://engineering.jhu.edu/news/engineering-meets-ballet-a-smart-solution-for-safer-pointe-work/> (Accessed: 18 September 2025).

EQUIPOINTE

**AN IMU-FSR BASED SYSTEM TO QUANTIFY
UNSAFE BALLET FOOTWORK**

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